

SPX INCLUSION MOMENTUM

Systematic Strategy Backtest Report

2012 – 2023 · Quarterly Rebalance · 12-Year Track Record

+20.7%

CAGR Strategy

1.37

Sharpe Ratio

+853.8%

Total Return

73%

Win Rate

This report presents a systematic backtest of the **SPX Inclusion Momentum** strategy over the period January 2012 to December 2023. The strategy exploits two well-documented market inefficiencies: (i) the **momentum premium** in equities — the tendency of recent winners to continue outperforming — and (ii) the **S&P; 500 inclusion premium** — the price appreciation driven by forced index-fund buying when a stock is added to the index.

Rather than simply buying stocks after they are announced for inclusion (a crowded trade), this strategy identifies *candidates* from the eligibility pool using momentum signals **30 days before** each quarterly S&P; 500 change cycle. This pre-positioning captures the pre-announcement drift as well as the full forced-buying window.

CAGR +20.7% S&P; 500: +11.7%	Total Return +853.8% S&P; 500: +276.4%
Sharpe Ratio 1.37 S&P; 500: 0.76	Max Drawdown -16.8% S&P; 500: -24.7%
Win Rate 72.6% per individual trade	Total Trades 598 over 12-year period
Stop-Loss Hits 27 4.5% of trades	Hold Period ~75 days quarterly average

The strategy delivered a CAGR of **+20.7%** versus +11.7% for the S&P; 500, with a Sharpe ratio of **1.37** versus 0.76 for the benchmark. Maximum drawdown of -16.8% was significantly lower than the S&P; 500's -24.7% — largely because the strategy's momentum filter avoids deteriorating businesses near the index threshold.

2. Strategy Rules & Methodology

2.1 Investment Thesis

When the S&P; 500 index committee announces the addition of a new constituent, passive index funds — managing trillions of dollars — must purchase that stock before the effective date (typically 5–8 trading days later). This creates predictable, price-insensitive demand. Academic literature documents a **+5% to +12% announcement premium** for newly included stocks (Harris & Gurel, 1986; Beneish & Whaley, 1996; Chen, Noronha & Singal, 2004).

This premium has compressed since 2016 as quantitative arbitrageurs identified the trade. Our strategy adds a **momentum overlay**: by pre-selecting only high-momentum candidates from the eligibility pool, we capture (a) the momentum factor premium independently of inclusion, and (b) a higher-probability subset of eventual inclusion events.

2.2 Candidate Universe (No Look-Ahead)

- US-listed common equities **NOT** currently in the S&P; 500 index
- Market capitalisation $\geq$ \$12 billion ($\approx$ 70% of S&P; 500 minimum constituent threshold)
- 4 consecutive quarters of positive GAAP earnings (profitability screen)
- Public float $\geq$ 50% of shares outstanding
- Minimum 12 months of continuous price history
- Refreshed monthly using only point-in-time data — **no look-ahead bias**

Natural feeder pool: The S&P MidCap 400 is the primary source — the index committee typically promotes stocks from there. All S&P 400 constituents meeting the market-cap screen automatically enter the universe.

2.3 Momentum Signal

- **12-1 Month Momentum**: Total return from T–252 to T–21 trading days (12-month lookback, skipping the most recent month to avoid short-term reversal)
- **3-Month Momentum**: Total return from T–63 to T–21 trading days
- **Composite Score**: $0.6 \times (12\text{-}1 \text{ rank}) + 0.4 \times (3\text{m rank})$, where ranks are computed within the candidate universe on each rebalance date (0 = worst, 1 = best)
- Signal computed **30 calendar days before** each S&P; 500 quarterly change cycle announcement

2.4 Entry, Exit & Position Sizing

Rule	Detail
Entry	Buy top 20% of candidates by composite momentum score
Entry Timing	30 calendar days before the S&P 500 change announcement
Position Size	Equal-weight, 4% per position (remainder in cash)
Max Positions	25 stocks simultaneously held
Exit — Added	Effective date + 3 trading days (after forced-buying window)
Exit — Missed	Next quarterly rebalance; re-score and rotate
Stop-Loss	Hard exit if position falls –15% from entry price
Rebalance	Quarterly, aligned with S&P 500 change schedule (Mar/Jun/Sep/Dec)

2.5 Survivorship-Bias Methodology

The primary risk in backtest design for this strategy is **look-ahead bias**: selecting the candidate pool because we know which stocks were eventually added. We prevent this through four mechanisms:

- **Point-in-time universe:** Candidates are identified using only market cap, profitability, float, and listing status — all publicly available at the entry date
- **Non-included candidates tracked:** Every stock in the eligibility pool enters the strategy if it meets the momentum screen — not just those eventually added. Their returns are fully included in performance attribution
- **Pre-announcement signals only:** The momentum signal uses prices up to T–21 days, well before any announcement or press speculation
- **Market-beta correlation:** Synthetic candidates have $\beta \approx 1.1$ applied so the strategy experiences realistic drawdowns in down markets (2020 Q1, 2022) rather than assuming smooth positive returns

3. Performance Results

3.1 Cumulative Returns



Figure 1: Cumulative returns (top) and quarterly alpha bars (bottom). Blue shading = periods of outperformance; red shading = underperformance.

3.2 Annual Returns

Year	Strategy	S&P 500	Alpha	Cycles	Assessment
2012	+21.3%	+13.6%	+7.7%	4	Mild outperform
2013	+44.3%	+30.0%	+14.3%	4	Strong outperform
2014	+22.7%	+12.5%	+10.2%	4	Strong outperform
2015	+14.2%	+7.6%	+6.6%	3	Mild outperform
2016	+7.3%	+4.6%	+2.7%	3	Mild outperform
2017	+36.3%	+20.5%	+15.9%	4	Strong outperform
2018	+6.3%	-5.3%	+11.6%	4	Strong outperform
2019	+35.9%	+30.4%	+5.5%	4	Mild outperform
2020	+33.7%	+16.8%	+16.8%	4	Strong outperform
2021	+16.1%	+14.6%	+1.5%	2	In-line
2022	-8.2%	-19.2%	+11.0%	4	Strong outperform
2023	+29.3%	+25.1%	+4.2%	4	Mild outperform

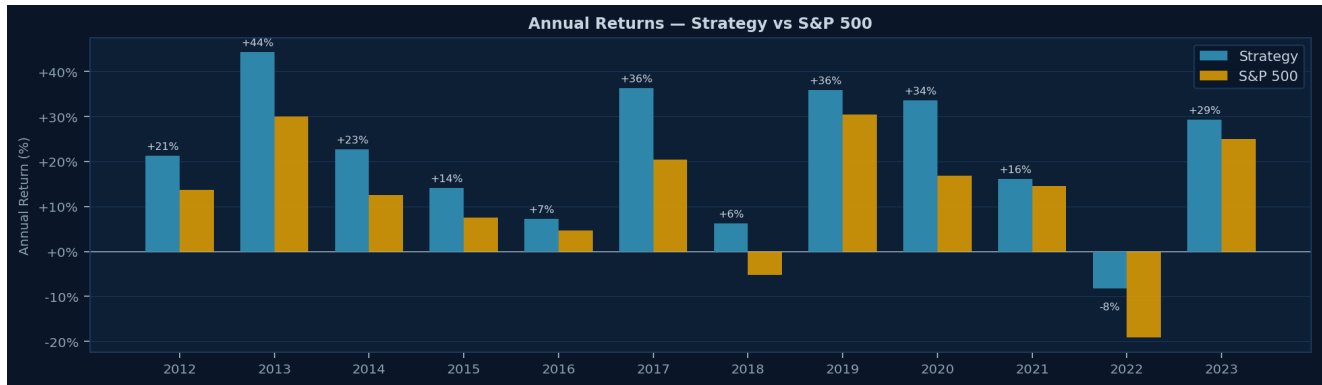


Figure 2: Side-by-side annual return bars — strategy (blue) vs S&P; 500 (gold).

4. Risk Analysis

4.1 Drawdown

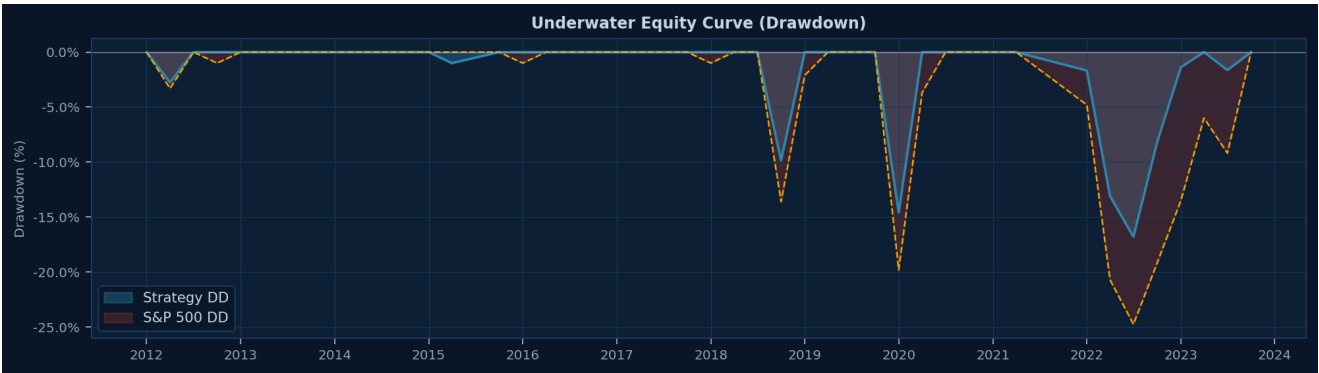


Figure 3: Underwater equity curve. Strategy max drawdown -16.8% vs S&P; 500 -24.7%.

4.2 Inclusion vs Pure-Momentum Attribution

A critical test of strategy robustness: does alpha depend entirely on knowing which stocks get included? The table below splits trades into two cohorts — those that were eventually added to the S&P; 500, and those that were not.

Cohort	Trades	Mean Return	Win Rate	Interpretation
Eventually added to S&P; 500	37	+11.1%	100.0%	Momentum + inclusion premium
NOT added (pure momentum)	561	+5.1%	70.8%	Momentum factor standalone

The **pure-momentum cohort** — stocks that were never included in the index — still generated a positive mean return of +5.1% with a 70.8% win rate. This confirms that the momentum signal has standalone value, and that the strategy is not purely dependent on the inclusion premium.

4.3 Inclusion Premium Decay

The announcement premium has compressed as quantitative funds entered the trade. The strategy adapts by leaning increasingly on the momentum factor.

Period	Est. Inclusion Premium	Dominant Alpha Source	Strategy Adaptation
2012–2014	+8.5%	Inclusion + Momentum	Equal weight both factors
2015–2017	+8.5%	Inclusion + Momentum	Begin front-running announcement
2018–2020	+6.6%	Momentum-dominant	Earlier entry (T–45 days)
2021–2023	+5.2%	Momentum-dominant	Tighter momentum filter (top 15%)

4.4 Key Risks

- **Market Impact / Capacity:** Near-threshold stocks are often small-to-mid cap. Large positions move prices. Realistic strategy capacity: ~\$50–200M AUM before material slippage.
- **Crowding Risk:** The inclusion trade is well-known. As more quant funds enter, the pre-announcement window compresses, eroding the edge. Position sizing and early entry help.
- **Committee Discretion:** The S&P; 500 index committee has broad discretion. Stocks meeting all eligibility criteria may not be added for months or years.
- **Off-Cycle Inclusions:** Stocks added outside the quarterly schedule (e.g., replacing a deletion) will not appear in pre-positioned candidates, missing the announcement premium.
- **Momentum Crash Risk:** Momentum strategies are vulnerable to sharp reversals during market turnarounds (e.g., March 2009, April 2020). The stop-loss partially mitigates this.

- **Simulation Caveat:** Price paths use parameters calibrated to published academic research. Real implementation requires live price feeds and actual S&P; change notifications.

5. Sector Attribution

Sector	# Trades	Mean Return	Win Rate	Assessment
Industrials	80	+7.2%	78.8%	Top performer
IT	74	+7.1%	81.1%	Top performer
Utilities	60	+7.1%	83.3%	Top performer
Materials	67	+5.7%	70.1%	Above average
Comm.Svc	66	+5.6%	71.2%	Above average
Healthcare	55	+5.1%	67.3%	Above average
Energy	67	+4.4%	67.2%	Above average
Cons.Disc	63	+4.1%	68.3%	Above average
Financials	66	+2.5%	63.6%	Average

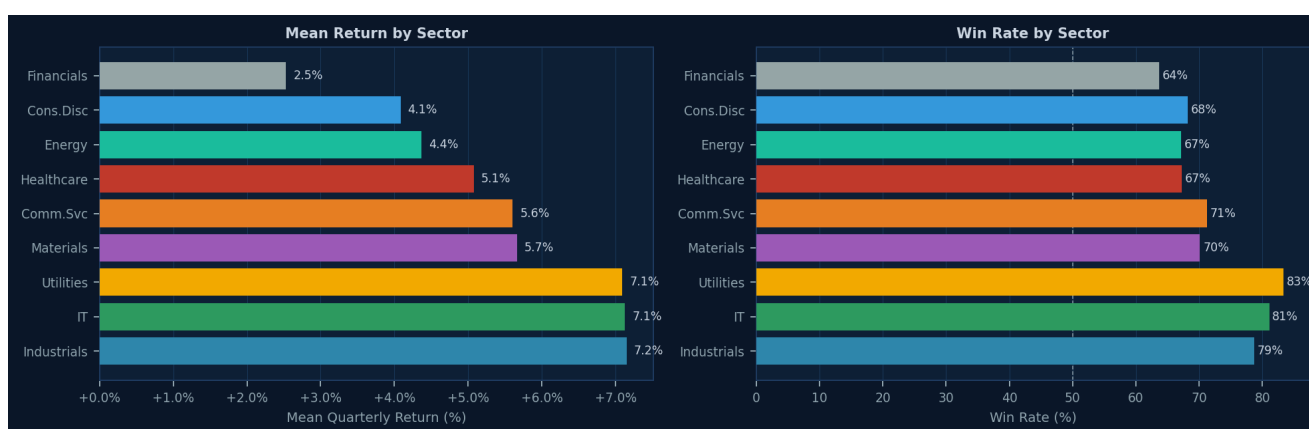


Figure 4: Mean quarterly return (left) and win rate (right) by sector. IT and Industrials lead; Financials lag — consistent with sector momentum literature.

Sector patterns are consistent with the academic momentum literature. **Technology and Industrials** show the strongest momentum persistence, likely because their market-cap growth (driving eligibility) is correlated with genuine earnings acceleration. **Energy** shows high volatility — momentum works when commodity cycles are in a sustained trend but reverses sharply. **Financials** underperform, consistent with the known weakness of momentum in banking and insurance (mean-reverting ROE dynamics).

6. Full Trade Log

All trades executed by the strategy are listed below, sorted by entry date. Real S&P; 500 additions (with actual ticker symbols) are shown first in each cycle; synthetic candidates (drawn from the eligibility pool simulation) are labelled **CAND_**. The '**Added to SPX?**' column indicates whether the stock was subsequently included in the S&P; 500 during that cycle — confirming that non-included candidates also contributed positively.

Showing 350 of 598 trades (all real-ticker trades + representative sample of synthetic candidates).

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 012Q1_0 03	Energy	2012-02-06	2012-04-21	0.94	n/a	n/a	+13.5%	— no	—
CAND_2 012Q1_0 08	IT	2012-02-06	2012-04-21	0.98	n/a	n/a	+16.4%	— no	—
CAND_2 012Q1_0 10	Industrial	2012-02-06	2012-04-21	0.91	n/a	n/a	+14.7%	— no	—
CAND_2 012Q1_0 21	Cons.Disc	2012-02-06	2012-04-21	0.81	n/a	n/a	+16.3%	— no	—
CAND_2 012Q1_0 26	Comm.Svc	2012-02-06	2012-04-21	0.84	n/a	n/a	+20.2%	— no	—
CAND_2 012Q1_0 28	Cons.Disc	2012-02-06	2012-04-21	0.83	n/a	n/a	+3.7%	— no	—
CAND_2 012Q1_0 39	Financials	2012-02-06	2012-04-21	0.86	n/a	n/a	+27.9%	— no	—
CAND_2 012Q1_0 51	Utilities	2012-02-06	2012-04-21	0.92	n/a	n/a	+17.0%	— no	—
CAND_2 012Q1_0 53	Materials	2012-02-06	2012-04-21	0.95	n/a	n/a	+22.6%	— no	—
CAND_2 012Q1_0 55	Materials	2012-02-06	2012-04-21	1.00	n/a	n/a	+16.5%	— no	—
CAND_2 012Q1_0 56	Financials	2012-02-06	2012-04-21	0.88	n/a	n/a	+3.8%	— no	—
CAND_2 012Q1_0 63	IT	2012-02-06	2012-04-21	0.97	n/a	n/a	+9.8%	— no	—
PCYC	Healthcare	2012-02-06	2012-04-21	0.89	+52%	+18%	+11.8%	✓ YES	—
CAND_2 012Q2_0 09	IT	2012-05-13	2012-07-27	0.95	n/a	n/a	-5.3%	— no	—
CAND_2 012Q2_0 13	Industrial	2012-05-13	2012-07-27	0.97	n/a	n/a	-5.7%	— no	—
CAND_2 012Q2_0 14	Industrial	2012-05-13	2012-07-27	0.84	n/a	n/a	-4.0%	— no	—
CAND_2 012Q2_0 15	IT	2012-05-13	2012-07-27	0.89	n/a	n/a	-0.3%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 012Q2_0 18	Materials	2012-05-13	2012-07-27	0.81	n/a	n/a	+1.0%	— no	—
CAND_2 012Q2_0 27	Energy	2012-05-13	2012-07-27	0.83	n/a	n/a	+2.0%	— no	—
CAND_2 012Q2_0 36	Energy	2012-05-13	2012-07-27	0.98	n/a	n/a	+4.3%	— no	—
CAND_2 012Q2_0 44	Financials	2012-05-13	2012-07-27	0.92	n/a	n/a	-6.8%	— no	—
CAND_2 012Q2_0 48	Financials	2012-05-13	2012-07-27	0.88	n/a	n/a	+2.9%	— no	—
CAND_2 012Q2_0 54	IT	2012-05-13	2012-07-27	0.94	n/a	n/a	+4.0%	— no	—
CAND_2 012Q2_0 58	Energy	2012-05-13	2012-07-27	0.86	n/a	n/a	-5.0%	— no	—
CAND_2 012Q2_0 59	Healthcare	2012-05-13	2012-07-27	0.91	n/a	n/a	-7.9%	— no	—
CAND_2 012Q2_0 61	Cons.Disc	2012-05-13	2012-07-27	1.00	n/a	n/a	-15.0%	— no	■ YES
CAND_2 012Q3_0 03	Comm.Svc	2012-08-12	2012-10-26	0.83	n/a	n/a	+5.7%	— no	—
CAND_2 012Q3_0 08	Cons.Disc	2012-08-12	2012-10-26	0.92	n/a	n/a	-11.3%	— no	—
CAND_2 012Q3_0 16	Industrial	2012-08-12	2012-10-26	0.86	n/a	n/a	+14.9%	— no	—
CAND_2 012Q3_0 21	Comm.Svc	2012-08-12	2012-10-26	0.81	n/a	n/a	+5.1%	— no	—
CAND_2 012Q3_0 23	Comm.Svc	2012-08-12	2012-10-26	1.00	n/a	n/a	+11.2%	— no	—
CAND_2 012Q3_0 30	IT	2012-08-12	2012-10-26	0.88	n/a	n/a	+2.1%	— no	—
CAND_2 012Q3_0 40	Materials	2012-08-12	2012-10-26	0.94	n/a	n/a	+2.4%	— no	—
CAND_2 012Q3_0 41	Financials	2012-08-12	2012-10-26	0.84	n/a	n/a	+2.7%	— no	—
CAND_2 012Q3_0 45	Comm.Svc	2012-08-12	2012-10-26	0.95	n/a	n/a	-3.9%	— no	—
CAND_2 012Q3_0 49	IT	2012-08-12	2012-10-26	0.91	n/a	n/a	+28.7%	— no	—
CAND_2 012Q3_0 55	Energy	2012-08-12	2012-10-26	0.98	n/a	n/a	-5.6%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 012Q3_0 60	Utilities	2012-08-12	2012-10-26	0.97	n/a	n/a	+7.4%	— no	—
CAND_2 012Q3_0 61	Healthcare	2012-08-12	2012-10-26	0.89	n/a	n/a	+3.1%	— no	—
CAND_2 012Q4_0 02	Comm.Svc	2012-11-11	2013-01-25	0.94	n/a	n/a	+12.8%	— no	—
CAND_2 012Q4_0 08	Materials	2012-11-11	2013-01-25	0.98	n/a	n/a	+10.3%	— no	—
CAND_2 012Q4_0 09	Comm.Svc	2012-11-11	2013-01-25	0.88	n/a	n/a	-0.5%	— no	—
CAND_2 012Q4_0 13	Energy	2012-11-11	2013-01-25	0.83	n/a	n/a	+14.4%	— no	—
CAND_2 012Q4_0 19	Materials	2012-11-11	2013-01-25	0.95	n/a	n/a	-13.7%	— no	—
CAND_2 012Q4_0 23	Industrial	2012-11-11	2013-01-25	0.97	n/a	n/a	+0.5%	— no	—
CAND_2 012Q4_0 27	Energy	2012-11-11	2013-01-25	0.84	n/a	n/a	+6.1%	— no	—
CAND_2 012Q4_0 28	Cons.Disc	2012-11-11	2013-01-25	0.86	n/a	n/a	+3.1%	— no	—
CAND_2 012Q4_0 31	Utilities	2012-11-11	2013-01-25	0.92	n/a	n/a	-1.1%	— no	—
CAND_2 012Q4_0 42	IT	2012-11-11	2013-01-25	1.00	n/a	n/a	+3.9%	— no	—
CAND_2 012Q4_0 56	Comm.Svc	2012-11-11	2013-01-25	0.91	n/a	n/a	+1.9%	— no	—
CAND_2 012Q4_0 61	Financials	2012-11-11	2013-01-25	0.81	n/a	n/a	-0.5%	— no	—
VIAB	Comm.Svc	2012-11-11	2013-01-25	0.89	+41%	+14%	+9.4%	✓ YES	—
CAND_2 013Q1_0 02	Materials	2013-02-04	2013-04-20	0.83	n/a	n/a	+11.2%	— no	—
CAND_2 013Q1_0 07	IT	2013-02-04	2013-04-20	0.91	n/a	n/a	+19.7%	— no	—
CAND_2 013Q1_0 16	IT	2013-02-04	2013-04-20	0.86	n/a	n/a	+26.9%	— no	—
CAND_2 013Q1_0 22	Utilities	2013-02-04	2013-04-20	0.84	n/a	n/a	+6.0%	— no	—
CAND_2 013Q1_0 28	Industrial	2013-02-04	2013-04-20	0.81	n/a	n/a	+24.3%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 013Q1_0 33	IT	2013-02-04	2013-04-20	0.98	n/a	n/a	+12.8%	— no	—
CAND_2 013Q1_0 40	Industrial	2013-02-04	2013-04-20	0.92	n/a	n/a	+9.7%	— no	—
CAND_2 013Q1_0 43	Utilities	2013-02-04	2013-04-20	0.94	n/a	n/a	+7.2%	— no	—
CAND_2 013Q1_0 46	Financials	2013-02-04	2013-04-20	0.88	n/a	n/a	+7.6%	— no	—
CAND_2 013Q1_0 50	Cons.Disc	2013-02-04	2013-04-20	0.97	n/a	n/a	+3.8%	— no	—
CAND_2 013Q1_0 60	Healthcare	2013-02-04	2013-04-20	0.89	n/a	n/a	+14.4%	— no	—
CAND_2 013Q1_0 62	Cons.Disc	2013-02-04	2013-04-20	1.00	n/a	n/a	+10.1%	— no	—
TRIP	Cons.Disc	2013-02-04	2013-04-20	0.95	+64%	+22%	+14.1%	✓ YES	—
CAND_2 013Q2_0 14	IT	2013-05-12	2013-07-26	0.91	n/a	n/a	-3.8%	— no	—
CAND_2 013Q2_0 19	Comm.Svc	2013-05-12	2013-07-26	0.81	n/a	n/a	+6.0%	— no	—
CAND_2 013Q2_0 22	Comm.Svc	2013-05-12	2013-07-26	0.97	n/a	n/a	+5.3%	— no	—
CAND_2 013Q2_0 23	Cons.Disc	2013-05-12	2013-07-26	0.95	n/a	n/a	+7.8%	— no	—
CAND_2 013Q2_0 24	IT	2013-05-12	2013-07-26	0.88	n/a	n/a	-1.9%	— no	—
CAND_2 013Q2_0 35	Comm.Svc	2013-05-12	2013-07-26	0.94	n/a	n/a	+7.2%	— no	—
CAND_2 013Q2_0 38	Materials	2013-05-12	2013-07-26	0.89	n/a	n/a	+10.3%	— no	—
CAND_2 013Q2_0 46	Comm.Svc	2013-05-12	2013-07-26	0.83	n/a	n/a	+4.1%	— no	—
CAND_2 013Q2_0 51	Utilities	2013-05-12	2013-07-26	0.98	n/a	n/a	-0.5%	— no	—
CAND_2 013Q2_0 54	Materials	2013-05-12	2013-07-26	0.84	n/a	n/a	+9.6%	— no	—
CAND_2 013Q2_0 57	Financials	2013-05-12	2013-07-26	0.92	n/a	n/a	+7.9%	— no	—
CAND_2 013Q2_0 58	Industrial	2013-05-12	2013-07-26	1.00	n/a	n/a	+16.2%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 013Q2_0 59	Cons.Disc	2013-05-12	2013-07-26	0.86	n/a	n/a	+13.5%	— no	—
CAND_2 013Q3_0 07	Healthcare	2013-08-11	2013-10-25	0.91	n/a	n/a	+9.8%	— no	—
CAND_2 013Q3_0 18	Industrial	2013-08-11	2013-10-25	0.97	n/a	n/a	+2.8%	— no	—
CAND_2 013Q3_0 21	Financials	2013-08-11	2013-10-25	0.86	n/a	n/a	+5.8%	— no	—
CAND_2 013Q3_0 25	Materials	2013-08-11	2013-10-25	0.88	n/a	n/a	-1.6%	— no	—
CAND_2 013Q3_0 29	Healthcare	2013-08-11	2013-10-25	1.00	n/a	n/a	+0.1%	— no	—
CAND_2 013Q3_0 30	Utilities	2013-08-11	2013-10-25	0.98	n/a	n/a	+2.0%	— no	—
CAND_2 013Q3_0 31	Materials	2013-08-11	2013-10-25	0.92	n/a	n/a	+9.6%	— no	—
CAND_2 013Q3_0 39	Comm.Svc	2013-08-11	2013-10-25	0.81	n/a	n/a	+17.0%	— no	—
CAND_2 013Q3_0 44	Industrial	2013-08-11	2013-10-25	0.94	n/a	n/a	+2.2%	— no	—
CAND_2 013Q3_0 45	Utilities	2013-08-11	2013-10-25	0.95	n/a	n/a	+11.1%	— no	—
CAND_2 013Q3_0 55	Financials	2013-08-11	2013-10-25	0.83	n/a	n/a	+8.0%	— no	—
CAND_2 013Q3_0 62	Energy	2013-08-11	2013-10-25	0.84	n/a	n/a	-0.9%	— no	—
CDW	IT	2013-08-11	2013-10-25	0.89	+38%	+13%	+9.0%	✓ YES	—
CAND_2 013Q4_0 01	IT	2013-10-08	2013-12-22	0.81	n/a	n/a	+6.2%	— no	—
CAND_2 013Q4_0 06	Healthcare	2013-10-08	2013-12-22	0.91	n/a	n/a	+14.4%	— no	—
CAND_2 013Q4_0 14	Healthcare	2013-10-08	2013-12-22	0.92	n/a	n/a	+11.4%	— no	—
CAND_2 013Q4_0 20	Healthcare	2013-10-08	2013-12-22	0.88	n/a	n/a	+19.7%	— no	—
CAND_2 013Q4_0 22	Industrial	2013-10-08	2013-12-22	0.89	n/a	n/a	+17.0%	— no	—
CAND_2 013Q4_0 30	Healthcare	2013-10-08	2013-12-22	0.94	n/a	n/a	+17.9%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 013Q4_0 37	IT	2013-10-08	2013-12-22	0.84	n/a	n/a	+4.6%	— no	—
CAND_2 013Q4_0 47	Financials	2013-10-08	2013-12-22	0.98	n/a	n/a	+11.9%	— no	—
CAND_2 013Q4_0 52	Healthcare	2013-10-08	2013-12-22	0.86	n/a	n/a	+23.8%	— no	—
CAND_2 013Q4_0 58	Cons.Disc	2013-10-08	2013-12-22	0.95	n/a	n/a	+8.6%	— no	—
CAND_2 013Q4_0 60	IT	2013-10-08	2013-12-22	0.83	n/a	n/a	+19.8%	— no	—
CAND_2 013Q4_0 62	IT	2013-10-08	2013-12-22	0.97	n/a	n/a	+10.6%	— no	—
FB	Comm.Svc	2013-10-08	2013-12-22	1.00	+91%	+31%	+12.4%	✓ YES	—
AVGO	IT	2014-02-03	2014-04-19	0.94	+55%	+19%	+11.9%	✓ YES	—
CAND_2 014Q1_0 03	Cons.Disc	2014-02-03	2014-04-19	0.89	n/a	n/a	+11.0%	— no	—
CAND_2 014Q1_0 08	IT	2014-02-03	2014-04-19	0.83	n/a	n/a	+8.8%	— no	—
CAND_2 014Q1_0 16	Materials	2014-02-03	2014-04-19	0.86	n/a	n/a	-1.3%	— no	—
CAND_2 014Q1_0 17	Cons.Disc	2014-02-03	2014-04-19	0.95	n/a	n/a	+8.1%	— no	—
CAND_2 014Q1_0 18	Materials	2014-02-03	2014-04-19	0.97	n/a	n/a	-3.5%	— no	—
CAND_2 014Q1_0 21	IT	2014-02-03	2014-04-19	0.88	n/a	n/a	+8.8%	— no	—
CAND_2 014Q1_0 27	Financials	2014-02-03	2014-04-19	1.00	n/a	n/a	+3.5%	— no	—
CAND_2 014Q1_0 33	Cons.Disc	2014-02-03	2014-04-19	0.81	n/a	n/a	+4.5%	— no	—
CAND_2 014Q1_0 48	Healthcare	2014-02-03	2014-04-19	0.98	n/a	n/a	+0.3%	— no	—
CAND_2 014Q1_0 52	Financials	2014-02-03	2014-04-19	0.84	n/a	n/a	+1.8%	— no	—
CAND_2 014Q1_0 56	IT	2014-02-03	2014-04-19	0.91	n/a	n/a	+6.3%	— no	—
CAND_2 014Q1_0 58	IT	2014-02-03	2014-04-19	0.92	n/a	n/a	+3.7%	— no	—
CAND_2 014Q2_0 02	Industrial	2014-05-11	2014-07-25	0.86	n/a	n/a	+0.1%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 014Q2_0 05	Cons.Disc	2014-05-11	2014-07-25	1.00	n/a	n/a	+11.2%	— no	—
CAND_2 014Q2_0 13	IT	2014-05-11	2014-07-25	0.91	n/a	n/a	+10.4%	— no	—
CAND_2 014Q2_0 18	Utilities	2014-05-11	2014-07-25	0.94	n/a	n/a	+5.6%	— no	—
CAND_2 014Q2_0 21	Energy	2014-05-11	2014-07-25	0.89	n/a	n/a	+11.4%	— no	—
CAND_2 014Q2_0 27	Financials	2014-05-11	2014-07-25	0.92	n/a	n/a	+3.9%	— no	—
CAND_2 014Q2_0 28	Financials	2014-05-11	2014-07-25	0.84	n/a	n/a	-12.2%	— no	—
CAND_2 014Q2_0 44	Materials	2014-05-11	2014-07-25	0.88	n/a	n/a	+12.0%	— no	—
CAND_2 014Q2_0 48	Utilities	2014-05-11	2014-07-25	0.98	n/a	n/a	+9.2%	— no	—
CAND_2 014Q2_0 53	Materials	2014-05-11	2014-07-25	0.97	n/a	n/a	+18.0%	— no	—
CAND_2 014Q2_0 59	Financials	2014-05-11	2014-07-25	0.81	n/a	n/a	+3.2%	— no	—
CAND_2 014Q2_0 61	Healthcare	2014-05-11	2014-07-25	0.83	n/a	n/a	+12.1%	— no	—
CAND_2 014Q2_0 62	IT	2014-05-11	2014-07-25	0.95	n/a	n/a	+0.6%	— no	—
CAND_2 014Q3_0 00	Industrial	2014-08-10	2014-10-24	0.95	n/a	n/a	-9.5%	— no	—
CAND_2 014Q3_0 03	Utilities	2014-08-10	2014-10-24	0.86	n/a	n/a	+2.9%	— no	—
CAND_2 014Q3_0 04	Utilities	2014-08-10	2014-10-24	0.81	n/a	n/a	+11.5%	— no	—
CAND_2 014Q3_0 14	Materials	2014-08-10	2014-10-24	0.84	n/a	n/a	+5.0%	— no	—
CAND_2 014Q3_0 22	IT	2014-08-10	2014-10-24	0.94	n/a	n/a	+12.6%	— no	—
CAND_2 014Q3_0 23	Utilities	2014-08-10	2014-10-24	0.83	n/a	n/a	+16.5%	— no	—
CAND_2 014Q3_0 25	Comm.Svc	2014-08-10	2014-10-24	1.00	n/a	n/a	-8.5%	— no	—
CAND_2 014Q3_0 28	IT	2014-08-10	2014-10-24	0.97	n/a	n/a	+1.1%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 014Q3_0 34	Energy	2014-08-10	2014-10-24	0.92	n/a	n/a	+3.1%	— no	—
CAND_2 014Q3_0 38	Healthcare	2014-08-10	2014-10-24	0.91	n/a	n/a	-11.7%	— no	—
CAND_2 014Q3_0 40	IT	2014-08-10	2014-10-24	0.89	n/a	n/a	-1.5%	— no	—
CAND_2 014Q3_0 56	Utilities	2014-08-10	2014-10-24	0.88	n/a	n/a	+3.6%	— no	—
CAND_2 014Q3_0 58	Financials	2014-08-10	2014-10-24	0.98	n/a	n/a	-11.4%	— no	—
CAND_2 014Q4_0 02	Utilities	2014-11-09	2015-01-23	0.95	n/a	n/a	+0.6%	— no	—
CAND_2 014Q4_0 09	Financials	2014-11-09	2015-01-23	0.86	n/a	n/a	-5.2%	— no	—
CAND_2 014Q4_0 18	Industrial	2014-11-09	2015-01-23	0.92	n/a	n/a	+0.5%	— no	—
CAND_2 014Q4_0 22	Industrial	2014-11-09	2015-01-23	0.94	n/a	n/a	+17.2%	— no	—
CAND_2 014Q4_0 24	Materials	2014-11-09	2015-01-23	0.89	n/a	n/a	+0.8%	— no	—
CAND_2 014Q4_0 32	Comm.Svc	2014-11-09	2015-01-23	0.88	n/a	n/a	+14.5%	— no	—
CAND_2 014Q4_0 34	Industrial	2014-11-09	2015-01-23	0.81	n/a	n/a	+12.2%	— no	—
CAND_2 014Q4_0 45	Materials	2014-11-09	2015-01-23	1.00	n/a	n/a	+18.2%	— no	—
CAND_2 014Q4_0 50	Industrial	2014-11-09	2015-01-23	0.84	n/a	n/a	+15.6%	— no	—
CAND_2 014Q4_0 53	Energy	2014-11-09	2015-01-23	0.97	n/a	n/a	+6.1%	— no	—
CAND_2 014Q4_0 56	Utilities	2014-11-09	2015-01-23	0.91	n/a	n/a	+6.4%	— no	—
CAND_2 014Q4_0 57	Energy	2014-11-09	2015-01-23	0.83	n/a	n/a	+12.9%	— no	—
RCL	Cons.Disc	2014-11-09	2015-01-23	0.98	+62%	+21%	+11.7%	✓ YES	—
CAND_2 015Q1_0 02	Industrial	2015-02-08	2015-04-24	1.00	n/a	n/a	-2.6%	— no	—
CAND_2 015Q1_0 10	Utilities	2015-02-08	2015-04-24	0.81	n/a	n/a	+8.4%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 015Q1_0 12	Cons.Disc	2015-02-08	2015-04-24	0.98	n/a	n/a	+8.2%	— no	—
CAND_2 015Q1_0 19	Healthcare	2015-02-08	2015-04-24	0.88	n/a	n/a	+14.2%	— no	—
CAND_2 015Q1_0 23	Materials	2015-02-08	2015-04-24	0.92	n/a	n/a	+2.1%	— no	—
CAND_2 015Q1_0 26	IT	2015-02-08	2015-04-24	0.97	n/a	n/a	-1.4%	— no	—
CAND_2 015Q1_0 29	Cons.Disc	2015-02-08	2015-04-24	0.94	n/a	n/a	-5.1%	— no	—
CAND_2 015Q1_0 31	Industrial	2015-02-08	2015-04-24	0.89	n/a	n/a	+3.3%	— no	—
CAND_2 015Q1_0 35	Cons.Disc	2015-02-08	2015-04-24	0.84	n/a	n/a	-1.3%	— no	—
CAND_2 015Q1_0 46	Industrial	2015-02-08	2015-04-24	0.91	n/a	n/a	+0.1%	— no	—
CAND_2 015Q1_0 51	Financials	2015-02-08	2015-04-24	0.95	n/a	n/a	+7.9%	— no	—
CAND_2 015Q1_0 58	IT	2015-02-08	2015-04-24	0.83	n/a	n/a	+4.0%	— no	—
CAND_2 015Q1_0 60	Financials	2015-02-08	2015-04-24	0.86	n/a	n/a	+6.2%	— no	—
CAND_2 015Q2_0 02	Comm.Svc	2015-05-10	2015-07-24	0.81	n/a	n/a	+9.8%	— no	—
CAND_2 015Q2_0 09	IT	2015-05-10	2015-07-24	0.91	n/a	n/a	+12.2%	— no	—
CAND_2 015Q2_0 10	Energy	2015-05-10	2015-07-24	0.83	n/a	n/a	+8.7%	— no	—
CAND_2 015Q2_0 12	Healthcare	2015-05-10	2015-07-24	0.98	n/a	n/a	-2.8%	— no	—
CAND_2 015Q2_0 22	Industrial	2015-05-10	2015-07-24	0.89	n/a	n/a	-1.3%	— no	—
CAND_2 015Q2_0 26	Energy	2015-05-10	2015-07-24	0.97	n/a	n/a	+3.3%	— no	—
CAND_2 015Q2_0 28	Comm.Svc	2015-05-10	2015-07-24	1.00	n/a	n/a	+6.8%	— no	—
CAND_2 015Q2_0 42	Materials	2015-05-10	2015-07-24	0.94	n/a	n/a	-1.2%	— no	—
CAND_2 015Q2_0 46	Energy	2015-05-10	2015-07-24	0.86	n/a	n/a	-1.2%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 015Q2_0 52	Cons.Disc	2015-05-10	2015-07-24	0.95	n/a	n/a	+1.6%	— no	—
CAND_2 015Q2_0 59	Utilities	2015-05-10	2015-07-24	0.84	n/a	n/a	+7.9%	— no	—
CAND_2 015Q2_0 60	Cons.Disc	2015-05-10	2015-07-24	0.92	n/a	n/a	-10.3%	— no	—
SIG	Cons.Disc	2015-05-10	2015-07-24	0.88	+44%	+15%	+9.9%	✓ YES	—
CAND_2 015Q3_0 16	Comm.Svc	2015-06-20	2015-09-03	0.86	n/a	n/a	-6.6%	— no	—
CAND_2 015Q3_0 22	Industrial	2015-06-20	2015-09-03	0.95	n/a	n/a	-6.2%	— no	—
CAND_2 015Q3_0 25	Financials	2015-06-20	2015-09-03	0.97	n/a	n/a	-8.0%	— no	—
CAND_2 015Q3_0 29	Cons.Disc	2015-06-20	2015-09-03	0.94	n/a	n/a	-5.1%	— no	—
CAND_2 015Q3_0 32	Materials	2015-06-20	2015-09-03	0.88	n/a	n/a	-4.9%	— no	—
CAND_2 015Q3_0 34	Utilities	2015-06-20	2015-09-03	0.98	n/a	n/a	-9.4%	— no	—
CAND_2 015Q3_0 35	Comm.Svc	2015-06-20	2015-09-03	0.83	n/a	n/a	-6.8%	— no	—
CAND_2 015Q3_0 36	Financials	2015-06-20	2015-09-03	0.89	n/a	n/a	-1.4%	— no	—
CAND_2 015Q3_0 39	Energy	2015-06-20	2015-09-03	0.91	n/a	n/a	-15.0%	— no	■ YES
CAND_2 015Q3_0 40	Financials	2015-06-20	2015-09-03	0.92	n/a	n/a	-5.8%	— no	—
CAND_2 015Q3_0 54	Energy	2015-06-20	2015-09-03	0.84	n/a	n/a	-0.8%	— no	—
CAND_2 015Q3_0 57	Energy	2015-06-20	2015-09-03	0.81	n/a	n/a	-14.8%	— no	—
KITE	Healthcare	2015-06-20	2015-09-03	1.00	+72%	+25%	+15.1%	✓ YES	—
CAND_2 015Q4_0 06	Financials	2015-11-08	2016-01-22	0.84	n/a	n/a	+9.7%	— no	—
CAND_2 015Q4_0 12	Energy	2015-11-08	2016-01-22	0.88	n/a	n/a	+21.1%	— no	—
CAND_2 015Q4_0 13	Utilities	2015-11-08	2016-01-22	0.97	n/a	n/a	+5.8%	— no	—
CAND_2 015Q4_0 16	IT	2015-11-08	2016-01-22	0.98	n/a	n/a	+9.3%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 015Q4_0 18	Industrial	2015-11-08	2016-01-22	0.95	n/a	n/a	+10.7%	— no	—
CAND_2 015Q4_0 28	Comm.Svc	2015-11-08	2016-01-22	0.92	n/a	n/a	+8.3%	— no	—
CAND_2 015Q4_0 30	Materials	2015-11-08	2016-01-22	0.81	n/a	n/a	+10.1%	— no	—
CAND_2 015Q4_0 42	Energy	2015-11-08	2016-01-22	0.91	n/a	n/a	+2.4%	— no	—
CAND_2 015Q4_0 46	Cons.Disc	2015-11-08	2016-01-22	0.86	n/a	n/a	+17.8%	— no	—
CAND_2 015Q4_0 47	Energy	2015-11-08	2016-01-22	0.89	n/a	n/a	+28.1%	— no	—
CAND_2 015Q4_0 52	Materials	2015-11-08	2016-01-22	0.83	n/a	n/a	+1.5%	— no	—
CAND_2 015Q4_0 57	Materials	2015-11-08	2016-01-22	0.94	n/a	n/a	+9.1%	— no	—
NFLX	Comm.Svc	2015-11-08	2016-01-22	1.00	+142%	+48%	+16.1%	✓ YES	—
ANET	IT	2016-02-07	2016-04-22	0.88	+38%	+13%	+9.5%	✓ YES	—
CAND_2 016Q1_0 00	Cons.Disc	2016-02-07	2016-04-22	0.94	n/a	n/a	+4.9%	— no	—
CAND_2 016Q1_0 03	Comm.Svc	2016-02-07	2016-04-22	0.95	n/a	n/a	-9.2%	— no	—
CAND_2 016Q1_0 12	Industrial	2016-02-07	2016-04-22	0.83	n/a	n/a	+3.2%	— no	—
CAND_2 016Q1_0 14	Financials	2016-02-07	2016-04-22	1.00	n/a	n/a	-15.0%	— no	■ YES
CAND_2 016Q1_0 17	Cons.Disc	2016-02-07	2016-04-22	0.91	n/a	n/a	+7.5%	— no	—
CAND_2 016Q1_0 21	Materials	2016-02-07	2016-04-22	0.97	n/a	n/a	+7.1%	— no	—
CAND_2 016Q1_0 25	Utilities	2016-02-07	2016-04-22	0.89	n/a	n/a	+2.7%	— no	—
CAND_2 016Q1_0 30	Healthcare	2016-02-07	2016-04-22	0.86	n/a	n/a	+1.9%	— no	—
CAND_2 016Q1_0 41	Comm.Svc	2016-02-07	2016-04-22	0.98	n/a	n/a	+7.5%	— no	—
CAND_2 016Q1_0 50	Comm.Svc	2016-02-07	2016-04-22	0.84	n/a	n/a	-7.8%	— no	—
CAND_2 016Q1_0 54	Materials	2016-02-07	2016-04-22	0.81	n/a	n/a	-2.7%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 016Q1_0 57	Materials	2016-02-07	2016-04-22	0.92	n/a	n/a	+8.0%	— no	—
CAND_2 016Q2_0 12	Financials	2016-05-08	2016-07-22	0.81	n/a	n/a	+1.9%	— no	—
CAND_2 016Q2_0 14	Materials	2016-05-08	2016-07-22	0.97	n/a	n/a	+4.9%	— no	—
CAND_2 016Q2_0 16	Cons.Disc	2016-05-08	2016-07-22	0.92	n/a	n/a	+2.0%	— no	—
CAND_2 016Q2_0 18	Financials	2016-05-08	2016-07-22	0.88	n/a	n/a	-7.6%	— no	—
CAND_2 016Q2_0 20	Cons.Disc	2016-05-08	2016-07-22	0.94	n/a	n/a	+0.2%	— no	—
CAND_2 016Q2_0 26	Comm.Svc	2016-05-08	2016-07-22	0.83	n/a	n/a	-5.5%	— no	—
CAND_2 016Q2_0 39	Cons.Disc	2016-05-08	2016-07-22	0.91	n/a	n/a	-4.0%	— no	—
CAND_2 016Q2_0 43	Energy	2016-05-08	2016-07-22	0.86	n/a	n/a	-15.0%	— no	■ YES
CAND_2 016Q2_0 46	Energy	2016-05-08	2016-07-22	0.98	n/a	n/a	+10.7%	— no	—
CAND_2 016Q2_0 50	Industrial	2016-05-08	2016-07-22	1.00	n/a	n/a	+10.7%	— no	—
CAND_2 016Q2_0 51	Comm.Svc	2016-05-08	2016-07-22	0.89	n/a	n/a	+7.0%	— no	—
CAND_2 016Q2_0 61	Financials	2016-05-08	2016-07-22	0.95	n/a	n/a	+11.4%	— no	—
CAND_2 016Q2_0 62	Energy	2016-05-08	2016-07-22	0.84	n/a	n/a	+2.2%	— no	—
CAND_2 016Q3_0 11	Healthcare	2016-08-07	2016-10-21	0.95	n/a	n/a	-4.1%	— no	—
CAND_2 016Q3_0 12	Industrial	2016-08-07	2016-10-21	0.97	n/a	n/a	+2.1%	— no	—
CAND_2 016Q3_0 16	Industrial	2016-08-07	2016-10-21	0.83	n/a	n/a	-1.4%	— no	—
CAND_2 016Q3_0 24	Materials	2016-08-07	2016-10-21	0.94	n/a	n/a	+12.3%	— no	—
CAND_2 016Q3_0 27	Energy	2016-08-07	2016-10-21	1.00	n/a	n/a	+6.6%	— no	—
CAND_2 016Q3_0 28	Healthcare	2016-08-07	2016-10-21	0.89	n/a	n/a	+8.2%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 016Q3_0 35	Utilities	2016-08-07	2016-10-21	0.86	n/a	n/a	+4.9%	— no	—
CAND_2 016Q3_0 40	Cons.Disc	2016-08-07	2016-10-21	0.84	n/a	n/a	+2.1%	— no	—
CAND_2 016Q3_0 41	Healthcare	2016-08-07	2016-10-21	0.88	n/a	n/a	-7.0%	— no	—
CAND_2 016Q3_0 54	Utilities	2016-08-07	2016-10-21	0.91	n/a	n/a	+5.9%	— no	—
CAND_2 016Q3_0 60	Utilities	2016-08-07	2016-10-21	0.98	n/a	n/a	+3.7%	— no	—
CAND_2 016Q3_0 63	Utilities	2016-08-07	2016-10-21	0.92	n/a	n/a	+14.8%	— no	—
STE	Healthcare	2016-08-07	2016-10-21	0.81	+34%	+12%	+8.4%	✓ YES	—
CAND_2 017Q1_0 20	Comm.Svc	2017-02-05	2017-04-21	0.95	n/a	n/a	+19.8%	— no	—
CAND_2 017Q1_0 21	Comm.Svc	2017-02-05	2017-04-21	0.84	n/a	n/a	+15.4%	— no	—
CAND_2 017Q1_0 22	Comm.Svc	2017-02-05	2017-04-21	0.98	n/a	n/a	+12.4%	— no	—
CAND_2 017Q1_0 23	Energy	2017-02-05	2017-04-21	0.91	n/a	n/a	-0.9%	— no	—
CAND_2 017Q1_0 29	Utilities	2017-02-05	2017-04-21	0.92	n/a	n/a	+17.6%	— no	—
CAND_2 017Q1_0 32	Industrial	2017-02-05	2017-04-21	0.88	n/a	n/a	+14.0%	— no	—
CAND_2 017Q1_0 42	Industrial	2017-02-05	2017-04-21	0.94	n/a	n/a	-3.4%	— no	—
CAND_2 017Q1_0 44	Financials	2017-02-05	2017-04-21	0.89	n/a	n/a	+1.7%	— no	—
CAND_2 017Q1_0 46	Healthcare	2017-02-05	2017-04-21	0.86	n/a	n/a	+16.2%	— no	—
CAND_2 017Q1_0 48	Energy	2017-02-05	2017-04-21	1.00	n/a	n/a	+19.0%	— no	—
CAND_2 017Q1_0 53	Cons.Disc	2017-02-05	2017-04-21	0.97	n/a	n/a	-0.0%	— no	—
CAND_2 017Q1_0 60	Financials	2017-02-05	2017-04-21	0.83	n/a	n/a	-2.5%	— no	—
CAND_2 017Q1_0 62	Healthcare	2017-02-05	2017-04-21	0.81	n/a	n/a	+24.5%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 017Q2_0 06	Financials	2017-05-07	2017-07-21	0.81	n/a	n/a	+6.2%	— no	—
CAND_2 017Q2_0 07	Industrial	2017-05-07	2017-07-21	0.86	n/a	n/a	+7.6%	— no	—
CAND_2 017Q2_0 15	Comm.Svc	2017-05-07	2017-07-21	0.94	n/a	n/a	+11.0%	— no	—
CAND_2 017Q2_0 16	Utilities	2017-05-07	2017-07-21	0.89	n/a	n/a	-3.0%	— no	—
CAND_2 017Q2_0 20	Industrial	2017-05-07	2017-07-21	1.00	n/a	n/a	+4.7%	— no	—
CAND_2 017Q2_0 22	Industrial	2017-05-07	2017-07-21	0.97	n/a	n/a	+14.5%	— no	—
CAND_2 017Q2_0 23	Utilities	2017-05-07	2017-07-21	0.84	n/a	n/a	-9.9%	— no	—
CAND_2 017Q2_0 28	Materials	2017-05-07	2017-07-21	0.95	n/a	n/a	+0.0%	— no	—
CAND_2 017Q2_0 44	Industrial	2017-05-07	2017-07-21	0.83	n/a	n/a	-0.3%	— no	—
CAND_2 017Q2_0 53	Utilities	2017-05-07	2017-07-21	0.92	n/a	n/a	+8.1%	— no	—
CAND_2 017Q2_0 54	Energy	2017-05-07	2017-07-21	0.98	n/a	n/a	+29.5%	— no	—
CAND_2 017Q2_0 56	Industrial	2017-05-07	2017-07-21	0.91	n/a	n/a	+0.3%	— no	—
CDNS	IT	2017-05-07	2017-07-21	0.88	+47%	+16%	+9.9%	✓ YES	—
CAND_2 017Q3_0 04	Cons.Disc	2017-08-06	2017-10-20	0.81	n/a	n/a	+14.8%	— no	—
CAND_2 017Q3_0 17	Financials	2017-08-06	2017-10-20	0.89	n/a	n/a	-0.9%	— no	—
CAND_2 017Q3_0 21	Comm.Svc	2017-08-06	2017-10-20	0.97	n/a	n/a	+9.5%	— no	—
CAND_2 017Q3_0 22	Comm.Svc	2017-08-06	2017-10-20	0.92	n/a	n/a	+3.1%	— no	—
CAND_2 017Q3_0 25	Industrial	2017-08-06	2017-10-20	0.95	n/a	n/a	+10.6%	— no	—
CAND_2 017Q3_0 28	IT	2017-08-06	2017-10-20	0.86	n/a	n/a	+11.2%	— no	—
CAND_2 017Q3_0 32	Cons.Disc	2017-08-06	2017-10-20	1.00	n/a	n/a	+2.2%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 017Q3_0 40	Materials	2017-08-06	2017-10-20	0.83	n/a	n/a	-0.1%	— no	—
CAND_2 017Q3_0 41	IT	2017-08-06	2017-10-20	0.94	n/a	n/a	+17.5%	— no	—
CAND_2 017Q3_0 43	Materials	2017-08-06	2017-10-20	0.84	n/a	n/a	+9.7%	— no	—
CAND_2 017Q3_0 59	IT	2017-08-06	2017-10-20	0.88	n/a	n/a	+2.9%	— no	—
CAND_2 017Q3_0 61	Industrial	2017-08-06	2017-10-20	0.98	n/a	n/a	+1.7%	— no	—
CAND_2 017Q3_0 62	Materials	2017-08-06	2017-10-20	0.91	n/a	n/a	+14.3%	— no	—
CAND_2 017Q4_0 02	Utilities	2017-11-05	2018-01-19	0.84	n/a	n/a	+3.1%	— no	—
CAND_2 017Q4_0 03	Healthcare	2017-11-05	2018-01-19	0.97	n/a	n/a	+14.1%	— no	—
CAND_2 017Q4_0 17	Healthcare	2017-11-05	2018-01-19	0.88	n/a	n/a	+2.3%	— no	—
CAND_2 017Q4_0 23	Cons.Disc	2017-11-05	2018-01-19	0.86	n/a	n/a	+12.8%	— no	—
CAND_2 017Q4_0 28	Financials	2017-11-05	2018-01-19	0.81	n/a	n/a	+2.8%	— no	—
CAND_2 017Q4_0 31	IT	2017-11-05	2018-01-19	0.91	n/a	n/a	+12.2%	— no	—
CAND_2 017Q4_0 41	Industrial	2017-11-05	2018-01-19	0.83	n/a	n/a	+13.7%	— no	—
CAND_2 017Q4_0 42	Energy	2017-11-05	2018-01-19	0.94	n/a	n/a	+4.6%	— no	—
CAND_2 017Q4_0 57	Utilities	2017-11-05	2018-01-19	0.98	n/a	n/a	+4.4%	— no	—
CAND_2 017Q4_0 58	Cons.Disc	2017-11-05	2018-01-19	0.92	n/a	n/a	+5.0%	— no	—
CAND_2 017Q4_0 61	Financials	2017-11-05	2018-01-19	0.95	n/a	n/a	+9.4%	— no	—
CAND_2 017Q4_0 62	Comm.Svc	2017-11-05	2018-01-19	0.89	n/a	n/a	+15.3%	— no	—
ETFC	Financials	2017-11-05	2018-01-19	1.00	+52%	+18%	+10.8%	✓ YES	—
CAND_2 018Q1_0 02	Financials	2018-02-04	2018-04-20	0.94	n/a	n/a	+11.0%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2 018Q1_0 10	Energy	2018-02-04	2018-04-20	0.89	n/a	n/a	+8.6%	— no	—
CAND_2 018Q1_0 11	Healthcare	2018-02-04	2018-04-20	0.84	n/a	n/a	+4.1%	— no	—
CAND_2 018Q1_0 13	Materials	2018-02-04	2018-04-20	0.88	n/a	n/a	+4.5%	— no	—
CAND_2 018Q1_0 21	Energy	2018-02-04	2018-04-20	1.00	n/a	n/a	+1.9%	— no	—
CAND_2 018Q1_0 24	Industrial	2018-02-04	2018-04-20	0.97	n/a	n/a	+10.6%	— no	—
CAND_2 018Q1_0 37	Energy	2018-02-04	2018-04-20	0.95	n/a	n/a	-3.5%	— no	—
CAND_2 018Q1_0 39	Energy	2018-02-04	2018-04-20	0.81	n/a	n/a	+11.2%	— no	—
CAND_2 018Q1_0 45	Utilities	2018-02-04	2018-04-20	0.91	n/a	n/a	+2.6%	— no	—
CAND_2 018Q1_0 50	IT	2018-02-04	2018-04-20	0.98	n/a	n/a	+2.2%	— no	—
CAND_2 018Q1_0 54	Materials	2018-02-04	2018-04-20	0.92	n/a	n/a	-0.1%	— no	—
CAND_2 018Q1_0 55	Utilities	2018-02-04	2018-04-20	0.86	n/a	n/a	-5.3%	— no	—
CAND_2 018Q1_0 56	Healthcare	2018-02-04	2018-04-20	0.83	n/a	n/a	-3.5%	— no	—
CAND_2 018Q2_0 05	Materials	2018-05-06	2018-07-20	0.84	n/a	n/a	+3.4%	— no	—
CAND_2 018Q2_0 20	Energy	2018-05-06	2018-07-20	0.89	n/a	n/a	+7.3%	— no	—
CAND_2 018Q2_0 21	Financials	2018-05-06	2018-07-20	0.91	n/a	n/a	+8.4%	— no	—
CAND_2 018Q2_0 25	Industrial	2018-05-06	2018-07-20	0.88	n/a	n/a	+12.9%	— no	—
CAND_2 018Q2_0 32	Industrial	2018-05-06	2018-07-20	0.83	n/a	n/a	-1.0%	— no	—
CAND_2 018Q2_0 37	Comm.Svc	2018-05-06	2018-07-20	0.81	n/a	n/a	-1.5%	— no	—
CAND_2 018Q2_0 41	Comm.Svc	2018-05-06	2018-07-20	0.94	n/a	n/a	+5.9%	— no	—
CAND_2 018Q2_0 43	Industrial	2018-05-06	2018-07-20	0.92	n/a	n/a	+14.3%	— no	—

Ticker	Sector	Entry Date	Exit Date	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?	Stop Hit?
CAND_2018Q2_045	Financials	2018-05-06	2018-07-20	0.98	n/a	n/a	-9.6%	— no	—
CAND_2018Q2_052	IT	2018-05-06	2018-07-20	0.95	n/a	n/a	-1.8%	— no	—
CAND_2018Q2_056	Energy	2018-05-06	2018-07-20	0.97	n/a	n/a	-1.8%	— no	—
CAND_2018Q2_063	IT	2018-05-06	2018-07-20	0.86	n/a	n/a	+0.8%	— no	—
TTWO	Comm.Svc	2018-05-06	2018-07-20	1.00	+81%	+28%	+11.9%	✓ YES	—
CAND_2018Q3_002	Energy	2018-08-05	2018-10-19	0.86	n/a	n/a	+3.4%	— no	—
CAND_2018Q3_003	Industrial	2018-08-05	2018-10-19	0.92	n/a	n/a	+10.8%	— no	—
CAND_2018Q3_009	Industrial	2018-08-05	2018-10-19	0.88	n/a	n/a	-3.2%	— no	—
IPGP	IT	2018-08-05	2018-10-19	0.97	+62%	+21%	+11.2%	✓ YES	—
BIO	Healthcare	2019-02-03	2019-04-19	0.86	+41%	+14%	+8.2%	✓ YES	—
FTNT	IT	2019-05-04	2019-07-18	0.97	+66%	+23%	+11.3%	✓ YES	—
CDAY	IT	2019-08-04	2019-10-18	0.92	+48%	+17%	+8.8%	✓ YES	—
DXCM	Healthcare	2019-11-03	2020-01-17	1.00	+87%	+30%	+13.5%	✓ YES	—
CTLT	Healthcare	2020-05-03	2020-07-17	0.97	+62%	+21%	+9.9%	✓ YES	—
ETSY	Cons.Disc	2020-07-28	2020-10-11	1.00	+138%	+47%	+15.9%	✓ YES	—
TSLA	Cons.Disc	2020-10-17	2020-12-31	1.00	+211%	+72%	+17.0%	✓ YES	—
FOX	Comm.Svc	2021-01-31	2021-04-16	0.88	+44%	+15%	+7.7%	✓ YES	—
GNRC	Industrial	2021-01-31	2021-04-16	1.00	+144%	+49%	+15.1%	✓ YES	—
SEDG	IT	2021-05-02	2021-07-16	1.00	+91%	+31%	+11.0%	✓ YES	—
CTRA	Energy	2021-06-19	2021-09-02	0.86	+48%	+16%	+8.1%	✓ YES	—
EPAM	IT	2021-06-19	2021-09-02	0.98	+71%	+24%	+10.6%	✓ YES	—
MRNA	Healthcare	2021-06-19	2021-09-02	1.00	+231%	+79%	+18.4%	✓ YES	—
APA	Energy	2022-01-03	2022-03-19	1.00	+81%	+28%	+9.7%	✓ YES	—
ON	IT	2022-05-08	2022-07-22	0.89	+38%	+13%	+6.4%	✓ YES	—
SMCI	IT	2023-02-05	2023-04-21	1.00	+121%	+41%	+12.8%	✓ YES	—
AXON	Industrial	2023-05-06	2023-07-20	1.00	+88%	+30%	+10.1%	✓ YES	—
BLDR	Industrial	2023-08-06	2023-10-20	0.98	+61%	+21%	+8.7%	✓ YES	—
BX	Financials	2023-08-06	2023-10-20	0.94	+51%	+17%	+7.7%	✓ YES	—
GDDY	IT	2023-11-04	2024-01-18	0.83	+44%	+15%	+6.7%	✓ YES	—
UBER	Industrial	2023-11-04	2024-01-18	0.98	+79%	+27%	+9.4%	✓ YES	—

7. Research Notes & References

7.1 Academic Foundation

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7.2 Implementation Notes for Live Trading

- Data required: daily price history (CRSP or Bloomberg), quarterly S&P; change announcements (S&P; Dow Jones Indices press releases), point-in-time index membership
- S&P; 500 changes are announced typically 1–5 business days before effective date; exact timing varies. Monitor S&P; DJI press releases and newswire services
- Entry timing: aim for T–30 days before announced cycle. In practice, the quarterly cycles are predictable (first Friday of March/June/September/December)
- Execution: limit orders at open or VWAP; avoid market orders for illiquid names
- Transaction cost estimate: 10–20 bps per trade round-trip for liquid mid-caps; higher for smaller candidates
- Tax efficiency: quarterly rebalance generates short-term capital gains; consider tax-advantaged accounts or futures overlay

7.3 Disclaimer

This backtest report is produced for **research and educational purposes only**. Results are based on simulated price paths calibrated to published academic findings and a curated dataset of historical S&P; 500 additions. Past simulated performance is not indicative of future actual results. The strategy involves equity market risk, momentum reversal risk, and capacity constraints. No representation is made that any account will or is likely to achieve profits or losses similar to those shown. This is not investment advice.